Practical Optimization - Lesson 5 : Constrained Optimization Homeworks

Vinh Thanh Ho, Paul Armand

EXERCISE 1

Consider the problem of finding the point on the parabola $x_2 = \frac{1}{5}(x_1-1)^2$ that is closest to $(x_1,x_2) = (1,2)$, in the Euclidean norm sense. The problem can be formulate as

$$\begin{array}{ll} \min & f(x) := (x_1-1)^2 + (x_2-2)^2 \\ \text{s.t.} & (x_1-1)^2 = 5x_2. \end{array}$$

Find all KKT points for this problem. Is the LICQ satisfied? Which of these points are solutions? By directly substituting the constraint into the objective function and eliminating the variable x_1, we obtain an unconstrained optimization problem. Show that the solutions to this problem cannot be solutions of the original problem.

EXERCISE 2

Consider the same problem as above, but the distance being measured in the Chebyshev sense, that is

$$\begin{array}{ll} \min & t \\ \text{s.t.} & (x_1-1)^2 = 5x_2 \\ & -t \le x_1 - 1 \le t \\ & -t \le x_2 - 2 \le t \end{array}$$

Solve the problem with AMPL and check the first order optimality conditions.

EXERCISE 3

Solve the following optimization problems. For each problem, state the first order optimality conditions and try to find the whole set of stationary points. Check the condition of constraint qualification. Check also the second order optimality conditions.

$$\begin{array}{ll} \min & -x_1 \\ \text{s.t.} & x_2 - x_1^3 - x_3^2 = 0 \\ & x_1^2 - x_2 - x_4^2 = 0 \end{array} \qquad \begin{array}{ll} \min & (x_1-1)^2 + (x_2 - x_1^2)^2 \\ \text{s.t.} & x_1 + x_3^2 = -1 \end{array}$$

$$\begin{array}{ll} \min & -x_1 \\ \text{s.t.} & x_2 - x_1^3 \ge 0 \\ & x_1^2 - x_2 \ge 0 \end{array} \qquad \begin{array}{ll} \min & \|x-a\|^2 \\ \text{s.t.} & x_1 \ge x_2 \ge \ldots \ge x_n \\ & \text{with } a = (1,2,3,\ldots,10) \end{array}$$

EXERCISE 4 The hanging chain problem

The problem is to find the shape of a hanging chain which is supported at its two ends and acted on only by its own weight. The shape will be found by solving a constrained optimization problem.
A chain of length $L > 0$ has m rigid links ($m > 1$), each of equal length L/m. Suppose that the two endpoints of the chain are fixed. Assuming that the mass of each link is concentrated on its middle, we consider the problem of computing the equilibrium position of the chain. The problem is described as follows. Let us denote by (x_k, y_k) the increment, horizontally and vertically, along the link (see Figure 1).

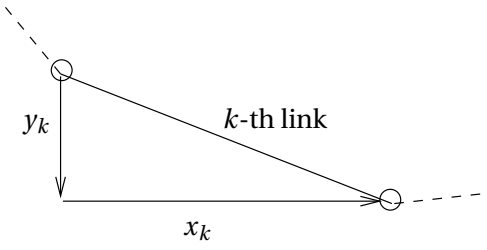


FIGURE 1 – Link coordinates

The links are numbered from 1 to m. We assume that one endpoint is fixed at (0,0). The position of the other endpoint is $\sum_{i=1}^{m}(x_i,y_i)$ and is constrained to be (a,b). The potential energy of the chain is given by

$$f(x,y) = \frac{1}{2}\sum_{k=1}^{m}\left(\sum_{i=1}^{k-1} y_i + \sum_{i=1}^{k} y_i\right) = c^\top y$$

where $c \in \mathbb{R}^m$ is defined by $c_k = m - k + \frac{1}{2}$ for $k = 1,\ldots,m$. The coordinate of the k-th link of the chain is defined by $\sum_{i=1}^{k}(x_i,y_i)$. The equilibrium positions can be calculated by minimizing the potential energy subject to involved physical constraints. This leads to the following optimization problem :

$$\begin{array}{ll} \min & c^\top y \\ \text{s.t.} & x_k^2 + y_k^2 = (L/m)^2, k = 1,\ldots,m \\ & \sum_{k=1}^{m} x_k = a \\ & \sum_{k=1}^{m} y_k = b. \end{array}$$

1. Write the first-order and second-order optimality conditions of this problem.
2. Write an AMPL model to solve the problem for a chain of length $L = 3$ with three links, $(a,b) = (1,0)$ and the starting points $x_k = y_k = 1$ for all $k = 1,\ldots,m$. Plot the equilibrium position of the chain. Check the first-order and second-order optimality conditions at the obtained solution.
3. We add an obstacle of the form

   $$x \mapsto he^{-\frac{(x-\mu)^2}{\sigma^2}} - v$$

   where x is the horizontal coordinate. Using AMPL, solve the problem with obstacle for a chain of length 3, 10 links, and last link fixed at (2,0), $h = 1.5$, $\mu = 1$, $\sigma^2 = 0.1$, $v = 2$, and the starting points $x_k = y_k = -1$ for all $k = 1,\ldots,m$. Complete your AMPL model to take into account the obstacle and give at least one optimal solution with your own choice of parameter values. Represent the optimal solution on a graphic (as Figure 2).

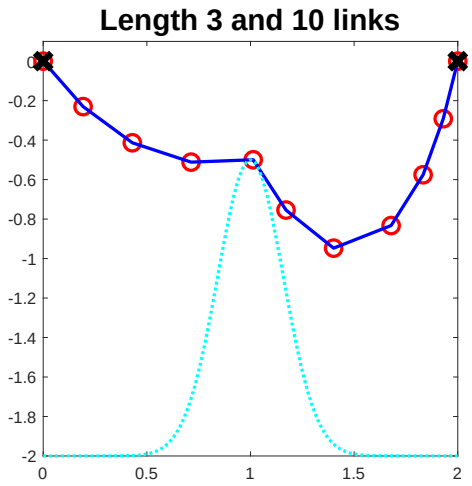


FIGURE 2 – Chain with obstacle.

Hint : it should discretize each link to compare with the obstacle.

```
# Links coordinates
var lx {k in 1..m} = sum {i in 1..k} x[i];
var ly {k in 1..m} = sum {i in 1..k} y[i];

# Number of discretization points of a link
param n := 20;

# Coordinate of the i-th point in the k-th link of the chain
var cx {k in 1..m-1,i in 0..n} = i/n*lx[k]+(1-i/n)*lx[k+1];
var cy {k in 1..m-1,i in 0..n} = i/n*ly[k]+(1-i/n)*ly[k+1];

# Obstacle
subject to obstacle {k in 1..m-1,i in 0..n}:
  cy[k,i] >= h*exp(-(cx[k,i]-mu)**2/sigmasquare) - nu;
```